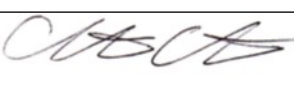


Guideline Title: Pediatric Bradycardia

Guideline Number: GUID-3203-PED005

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** Ensure prompt identification and intervention to restore adequate cardiac output and a stable cardiac rhythm.
- II. **DEFINITIONS:**
- III. **DEPARTMENT GUIDELINE:**
 - A. Immediately assess for hypoxia, hypoperfusion, hypotension, hypothermia, altered mental status, & signs of shock.
 - B. If HR is < 60 bpm with poor perfusion despite oxygenation and ventilation begin chest compressions
 - C. If the patient is unstable, begin to resuscitate. If the patient is stable, attempt to identify the cause of the bradycardia and resolve it.
 - D. Manage the airway aggressively with high FiO₂ oxygen and ventilatory support with positive pressure ventilation. Intubate, if necessary, per *GUID3203-G008 Rapid Sequence Induction for Endotracheal Intubation* and *GUID3203-PR020 Endotracheal Intubation*. Apply ETCO₂ monitor.
 - E. Apply cardiac monitor, monitor pulse, blood pressure, and oximetry.
 - F. If there are no signs of cardiopulmonary compromise, observe and perform frequent reassessments, obtain a 12-lead ECG, and treat underlying causes
 - G. If from a sending facility obtain appropriate laboratory studies (potassium, glucose, ionized calcium, magnesium, blood gas for pH, toxicology screen.
 - H. For symptomatic bradycardia that persists despite effective oxygenation and ventilation
 - i. **Epinephrine (1:10,000) 0.01 mg/kg (0.1 ml/kg) IV/IO**, max dose 1 mg.
 - ii. **Epinephrine (1:1,000) 0.1 mg/kg (0.1 ml/kg) via ETT**, max dose 1 mg
 - iii. Re-dose every 3-5 minutes.
 - I. For increased vagal tone or primary AV block consider:
 - i. **Atropine 0.02 mg/kg IV/IO**, Min dose 0.1 mg, Max dose 0.5 mg.
 - ii. **Atropine 0.04-0.06 mg/kg via ETT**
 - iii. May repeat once in 5 minutes. Larger doses may be required for organophosphate poisoning.
 - J. For bradycardia caused by a complete heart block or abnormal sinus node function or post-surgical correction of congenital disease consider Transcutaneous pacing per *GUID3203-PR022 - Transcutaneous External Pacing (TCP)*.
 - i. For TCP, provide sedation and/or analgesia if permissible per *GUID3203-G005 - Analgesia and Sedation Management*.

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K. Consider underlying causes:

- i. Hypoxia: high concentrated O₂ with assisted ventilations
- ii. H⁺ ions – Acidosis: Provide ventilation to treat respiratory acidosis secondary to hypercarbia. Consider Sodium Bicarb in severe metabolic acidosis with expert opinion
- iii. Hyperkalemia: with expert consultation restore normal potassium concentration by stopping fluids that contain potassium. Consider Calcium chloride or gluconate, albuterol, sodium bicarb, insulin with glucose administration, loop diuretics, kayexalate.
- iv. Hypothermia: warm the child but avoid hyperthermia
- v. Heart block: See above for atropine or TCP
- vi. Toxic ingestion: treat per *GUID3203-M019 Overdose, Poisoning, & Toxic Exposure*
- vii. Head Trauma: bradycardia with head trauma is an ominous sign of high intracranial pressure. Provide oxygenation, ventilation, and consider a brief period of mild hyperventilation

L. Always consider the latest recommendations from the American Heart Association.

IV. **DOCUMENTATION:** EMS Charts

V. **REFERENCES:**

- A. Pediatric Advanced Life Support Manual, 2020. American Heart Association.